1). Define Flutter and write the steps to install Flutter SDK in Ubuntu.
2). Write a "Hello World" program using Stateless and Stateful widgets.
3).Write an example program to demonstrate decision-making statements in Dart.
4). Write an example program to demonstrate looping statements in Dart.
5). Write an example program to demonstrate inheritance in Dart.
6). Demonstrate how to implement functions in Dart.
7).Write a Dart program to implement the RichText widget.
8). Write a Dart program to display a popup menu button widget.
9). Write a Dart program to demonstrate a TextField widget.
10).Write a Dart program to insert an image in an application window.
11). Write a Dart program to implement the Container widget.
12).Write a Dart program to demonstrate the Floating Action Button widget.

2) Write a "Hello World" program using Stateless and Stateful widgets.

STATELESS WIDGET:

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: const Text('Hello World with Stateless Widget'),
        ),
        body: const Center(
          child: Text(
            'Hello, World!',
            style: TextStyle(fontSize: 24),
          ),
        ),
      ),
    );
  }
}
```

STATEFULL WIDGET:

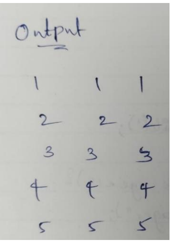
```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: HelloWorldStateful(),
    );
  }
}
class HelloWorldStateful extends StatefulWidget {
  @override
  State<HelloWorldStateful> createState() =>
_HelloWorldStatefulState();
}
class _HelloWorldStatefulState extends
State<HelloWorldStateful> {
  String message = 'Hello, World!';
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Hello World with Stateful Widget'),
      ),
      body: Center(
        child: Text(
          message,
          style: const TextStyle(fontSize: 24),
        ),
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          setState(() {
            message = 'Hello, Flutter!';
          });
        },
        child: const Icon(Icons.refresh),
      ),
    );
  }
}
```

3).Write an example program to demonstrate decision-making statements in Dart.
3.decision making statements

```
void main() {
  int age = 20;
  if (age < 13) {
    print("You are a Child");
  } else if (age >= 13 && age < 18) {
    print("You are a Teenager");
  } else if (age >= 18 && age < 60) {
    print("You are an Adult");
  } else {
    print("You are a Senior");
  }
}
```

4). Write an example program to demonstrate looping statements in Dart.
4.looping statements

```
void main() {
  // For loop
  for (int i = 1; i <= 5; i++) {
    print(i);
  }
  // While loop
  int j = 1;
  while (j <= 5) {
    print(j);
    j++;
  }
  // Do-while loop
  int k = 1;
  do {
    print(k);
    k++;
  } while (k <= 5);
}
```



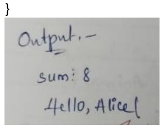
5). Write an example program to demonstrate inheritance in Dart.
5. Inheritance

```
class Animal {
  void eat() {
    print("Animal is eating");
  }
}

class Dog extends Animal {
  void bark() {
    print("Dog is barking");
  }
}
void main() {
  var dog = Dog();
  dog.eat();
  dog.bark();
}
```

OUTPUT: Animal is eating
dog is barking

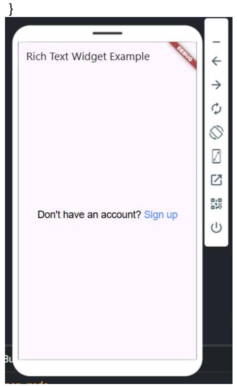
6). Demonstrate how to implement functions in Dart.

```
int add(int a, int b) {
  return a + b;
}

void greet(String name) {
  print("Hello, $name");
}

void main() {
  int sum = add(3, 5);
  print("Sum: $sum");
  greet("Alice");
}
```



7).Write a Dart program to implement the RichText widget.

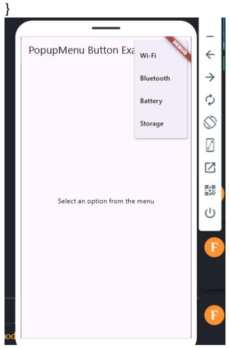
```
import 'package:flutter/material.dart';
import 'package:flutter/gestures.dart';
void main()
{
runApp(MyApp());
}
 class MyApp extends StatelessWidget
 { @override
 Widget build(BuildContext context)
{
 return MaterialApp(
theme: ThemeData(
primarySwatch: Colors.green,
),
home: MyTextPage()
 );
}
}
class MyTextPage extends StatelessWidget
{ @override
Widget build(BuildContext context)
{ return Scaffold(
appBar: AppBar(
 title:Text("Rich Text Widget Example")
),
 body: Container(
child: Center(
child: RichText(
 text: TextSpan(
 text: 'Don\'t have an account?',
 style: TextStyle(color: Colors.black, fontSize: 20),
children: <TEXTSPAN>[
TextSpan(text: ' Sign up',
style: TextStyle(color: Colors.blueAccent, fontSize: 20),
recognizer: TapGestureRecognizer()
..onTap = () {}
)
]
 ),
 ),
 )
 )
 );
 }
}
```



8). Write a Dart program to display a popup menu button widget.

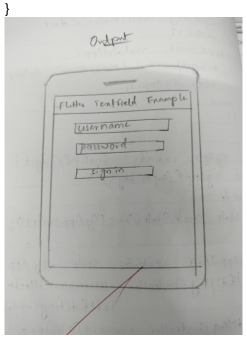
```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
```

```
        title: const Text('Popup Menu Button Example'),
        actions: <Widget>[
          PopupMenuButton<Choice>(
            onSelected: (Choice choice) {
              showDialog(
                context: context,
                builder: (context) => AlertDialog(
                  content: Column(
                    mainAxisSize: MainAxisSize.min,
                    children: [
                      Icon(choice.icon, size: 100),
                      Text(choice.name),
                    ],
                  ),
                ),
              );
            },
            itemBuilder: (context) {
              return choices.map((Choice choice) {
                return PopupMenuItem<Choice>(
                  value: choice,
                  child: Text(choice.name),
                );
              }).toList();
            },
          ),
        ],
      ),
      body: const Center(
        child: Text('Select an option from the menu'),
      ),
    ),
  );
 }
}
class Choice {
  const Choice({required this.name, required this.icon});
  final String name;
  final IconData icon;
}
const List<Choice> choices = <Choice>[
  Choice(name: 'Wi-Fi', icon: Icons.wifi),
  Choice(name: 'Bluetooth', icon: Icons.bluetooth),
  Choice(name: 'Battery', icon: Icons.battery_alert),
  Choice(name: 'Storage', icon: Icons.storage),
];
}
```



9). Write a Dart program to demonstrate a TextField widget.

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MaterialApp(
    debugShowCheckedModeBanner: false,
    home: MyApp(),
  ));
}
class MyApp extends StatefulWidget {
  const MyApp({super.key});

  @override
  MyAppState createState() => MyAppState();
}
class MyAppState extends State<MyApp> {
  final TextEditingController usernameController =
TextEditingController();
  final TextEditingController passwordController =
TextEditingController();
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Flutter TextField Example'),
        backgroundColor: Colors.blue,
      ),
      body: Padding(
        padding: const EdgeInsets.all(15),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.stretch,
          children: <Widget>[
            TextField(
              controller: usernameController,
              decoration: const InputDecoration(
                border: OutlineInputBorder(),
                labelText: 'Username',
                hintText: 'Enter your username',
              ),
            ),
            const SizedBox(height: 30),
            TextField(
              controller: passwordController,
              obscureText: true,
              decoration: const InputDecoration(
                border: OutlineInputBorder(),
                labelText: 'Password',
                hintText: 'Enter your password',
              ),
            ),
            const SizedBox(height: 30),
            ElevatedButton(
              style: ElevatedButton.styleFrom(
                backgroundColor: Colors.blue,
                foregroundColor: Colors.white,
                padding: const EdgeInsets.symmetric(horizontal: 30,
vertical: 15),
              ),
              onPressed: () {
                final user = usernameController.text;
                final pass = passwordController.text;

                ScaffoldMessenger.of(context).showSnackBar(
                  SnackBar(
                    content: Text('Username: $user | Password: $pass'),
                  ),
                );
              },
              child: const Text('Sign In'),
            ),
          ],
        ),
      ),
    );
  }
}
```



10).Write a Dart program to insert an image in an application window.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: const Text('Load Image from Internet'),
        ),
        body: Center(
          child: Image.network(
            'https://flutter.github.io/assets-for-api-
docs/assets/widgets/owl.jpg',
            width: 250,
            height: 250,
            fit: BoxFit.cover,
          ),
        ),
      ),
    );
  }
}
```

11). Write a Dart program to implement the Container widget.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: const Text('Container Widget Example'),
        ),
        body: Center(
          child: Container(
            width: 200,
            height: 200,
            alignment: Alignment.center,
            decoration: BoxDecoration(
              color: Colors.blueAccent,
              borderRadius: BorderRadius.circular(20),
              border: Border.all(
                color: Colors.black,
                width: 3,
              ),
              boxShadow: const [
                BoxShadow(
                  color: Colors.grey,
                  blurRadius: 8,
                  offset: Offset(4, 4),
                ),
              ],
            ),
            child: const Text(
              'Hello, Flutter!',
              style: TextStyle(
                fontSize: 20,
                color: Colors.white,
                fontWeight: FontWeight.bold,
              ),
            ),
          ),
        ),
      ),
    );} }
```

12).Write a Dart program to demonstrate the Floating Action Button widget.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: FloatingButtonExample(),
    );
  }
}
class FloatingButtonExample extends StatefulWidget {
  @override
  State<FloatingButtonExample> createState() =>
_FloatingButtonExampleState();
}
class _FloatingButtonExampleState extends
State<FloatingButtonExample> {
  int counter = 0;
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Floating Action Button Example'),
      ),
      body: Center(
        child: Text(
          'Button pressed $counter times',
          style: const TextStyle(fontSize: 22),
        ),
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          setState(() {
            counter++;
          });
        },
        tooltip: 'Increment',
        child: const Icon(Icons.add),
        backgroundColor: Colors.blue,
```

```
    ),
   );
  }
}
```